

EchoChain is a data analytics system that tracks what happens to a product after it is sold.

For example, a company manufactures a laptop.

Normally, the company knows:

Manufacturing → Sale → Customer

But after the customer buys it, the company may not know.

EchoChain collects this information and analyzes the complete lifecycle of the product.

--->**A data blind spot:**

An area where the company does not have enough information or visibility.

Manufacturing → Sale → ??? → Resale / Repair / Waste

↑

DATA BLIND SPOT

EchoChain combines:

Internal company data

Product/SKU

BOM

Warranty

Component failures

with

Secondary-market data

eBay listings

Resale price

Product condition

Component value

--->**Environmental impact:**

The effect that a product has on the environment.

Suppose a laptop is thrown away even though its display and RAM are still working.

That creates unnecessary waste.

The company can reduce waste and the need to manufacture completely new components.

---->Landfill diversion:

keeping waste away from landfills.

A landfill is a place where discarded waste is permanently dumped.

---->Potential for Profitable Refurbishment

Can we take an old/used product, repair it, and sell it again for more money than it costs to recover and repair it?

----->Circularity Score:

measures how well a product can remain useful instead of becoming waste.

- A low-circularity product might look like:

Manufacture → Sell → Use → Throw away

- A high-circularity product might look like:

Manufacture→ Sell→ Use→ Repair→ Refurbish→ Resell→ Reuse

So the Circularity Score is an analytics KPI that helps the sustainability executive identify products with strong opportunities for reuse and recovery.

---->A buy-back program:

The manufacturer offers customers money or credit to return their old products.

--->Refurbishment:

means taking a used product, repairing/replacing necessary parts, testing it, and preparing it for another user.

MANUFACTURER



Product / BOM Data



Warranty Data



What components fail?



SECONDARY MARKET

eBay data



Resale price + condition



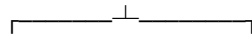
What still has value?



Combine the data



Circularity Analysis



Environmental



Financial

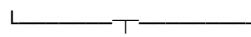
Impact

Value



Landfill diversion

Refurbishment opportunity



CIRCULARITY SCORE



Business Decision



Buy-back Program



Refurbishment



Resale